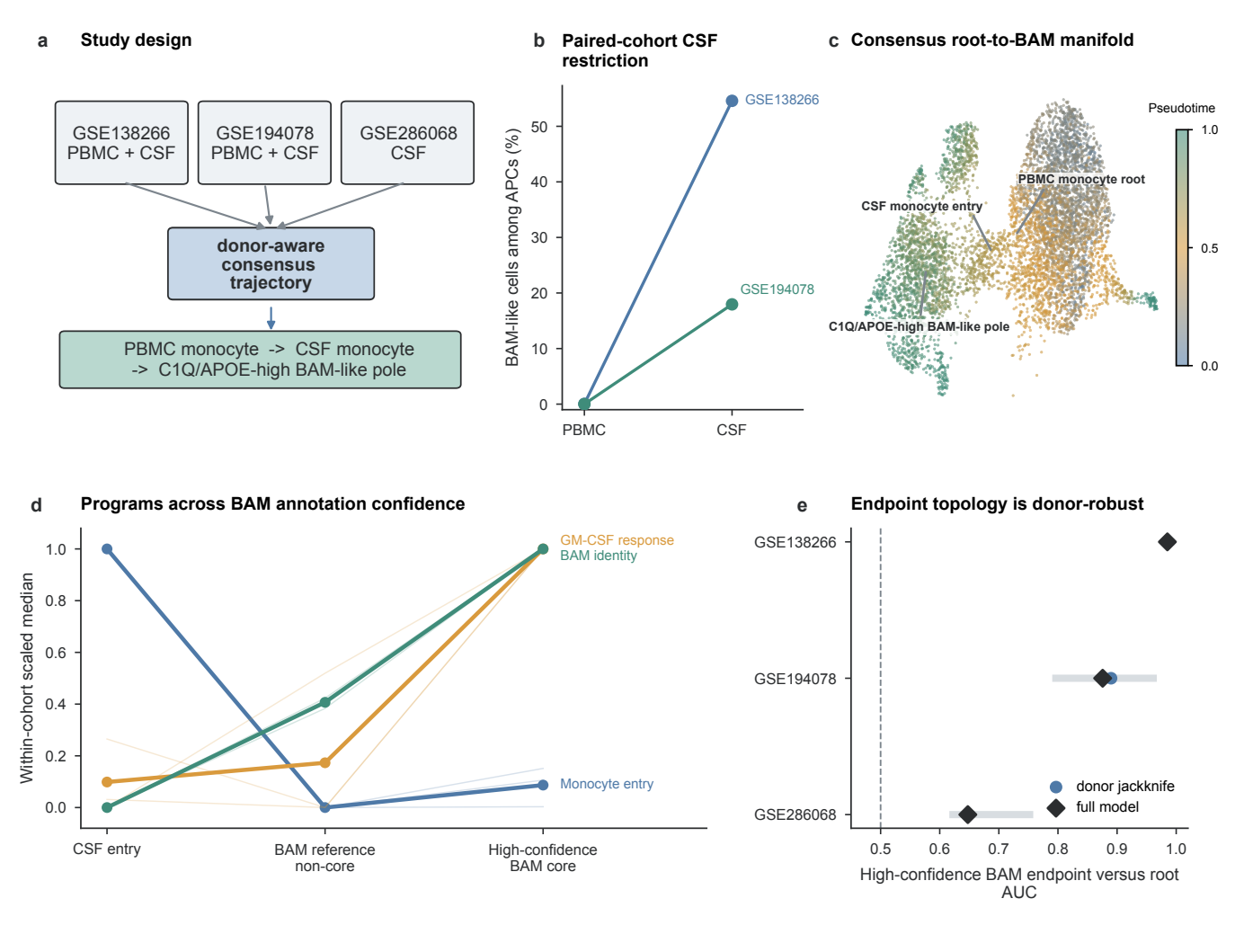

a
Study design
GSE138266
PBMC + CSF
GSE194078
PBMC + CSF
GSE286068
CSF
donor-aware
consensus
trajectory
PBMC monocyte -> CSF monocyte
-> C1Q/APOE-high BAM-like pole
b
Paired-cohort CSF
restriction
BAM-like cells among APCs (%)
0
10
20
30
40
50
GSE138266
GSE194078
PBMC
CSF
c
Consensus root-to-BAM manifold
Pseudotime
1.0
0.5
0.0
PBMC monocyte root
CSF monocyte entry
C1Q/APOE-high BAM-like pole
d
Programs across BAM annotation confidence
Within-cohort scaled median
0.0
0.2
0.4
0.6
0.8
1.0
GM-CSF response
BAM identity
Monocyte entry
CSF entry
BAM reference
non-core
High-confidence
BAM core
e
Endpoint topology is donor-robust
GSE138266
GSE194078
GSE286068
donor jackknife
full model
0.5
0.6
0.7
0.8
0.9
1.0
High-confidence BAM endpoint versus root
AUC